



NSF REU AI-Empowered Cybersecurity

# Behavioral Authentication Under Attack: AI Mouse Bots, Human Impostors, and User-Specific Vulnerability

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## Introduction

You can recognize a user by how they behave — the way they move a mouse, pause, and scroll is distinctive enough to identify you. Behavioral authentication uses this to continuously verify who's really behind the screen, catching an impostor even after they've already gotten past a password. But AI-generated impostors are getting better at mimicking real behavior, from simple rule-based bots to models trained specifically to copy how a real person moves. This kind of defense is already deployed where separating a real user from a bot has real stakes:

- Financial** platforms protecting against account takeover
- Government** systems safeguarding sensitive access
- Game anti-cheat** catching bots and cheaters

## Research Questions

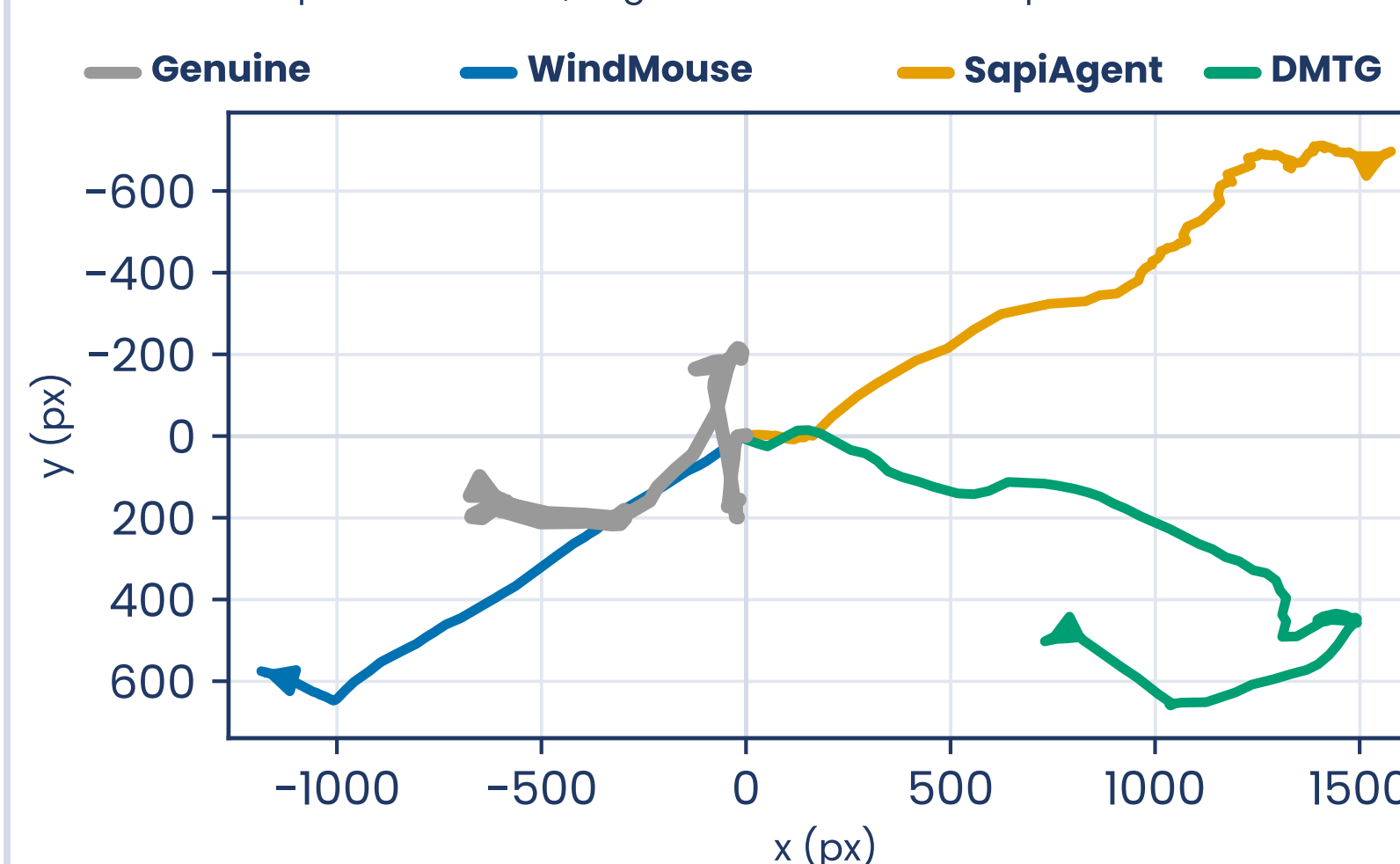
- How does attack success change as mouse bots become more sophisticated, and how do they compare with real human impostors?
- Are some users more vulnerable to behavioral imitation than others, and does the strongest attacker depend on the target user?
- Can additional defense mechanisms improve robustness against synthetic attacks without significantly reducing legitimate-user acceptance?

## Datasets & Environments

- Structured time-series (Balabit Mouse Dynamics Challenge)** — Timestamped (x, y) cursor positions per session from 10 enrolled users, recorded from real sessions and analyzed offline.
- Real and synthetic attackers** — Real human-impostor sessions set the human baseline. Three mouse bots (WindMouse, SapiAgent, DMTG) run against every user under identical conditions, at a scale real impostors cannot match.
- Environment** — Attackers built in Colab, except the diffusion attacker, trained locally in PyTorch on an RTX 5070. Defenders scored in Colab.

### Example: real user vs. mouse bots

One 128-step window each, aligned to a shared start point



This is an example of what a real session actually looks like: paths are shifted to a common start point so shape (not screen position) can be compared directly. Genuine is a real user; WindMouse, SapiAgent, and DMTG are mouse bots trying to mimic it.

Figure 1. Example genuine vs. synthetic cursor paths, one 128-step window each.

## Methodology

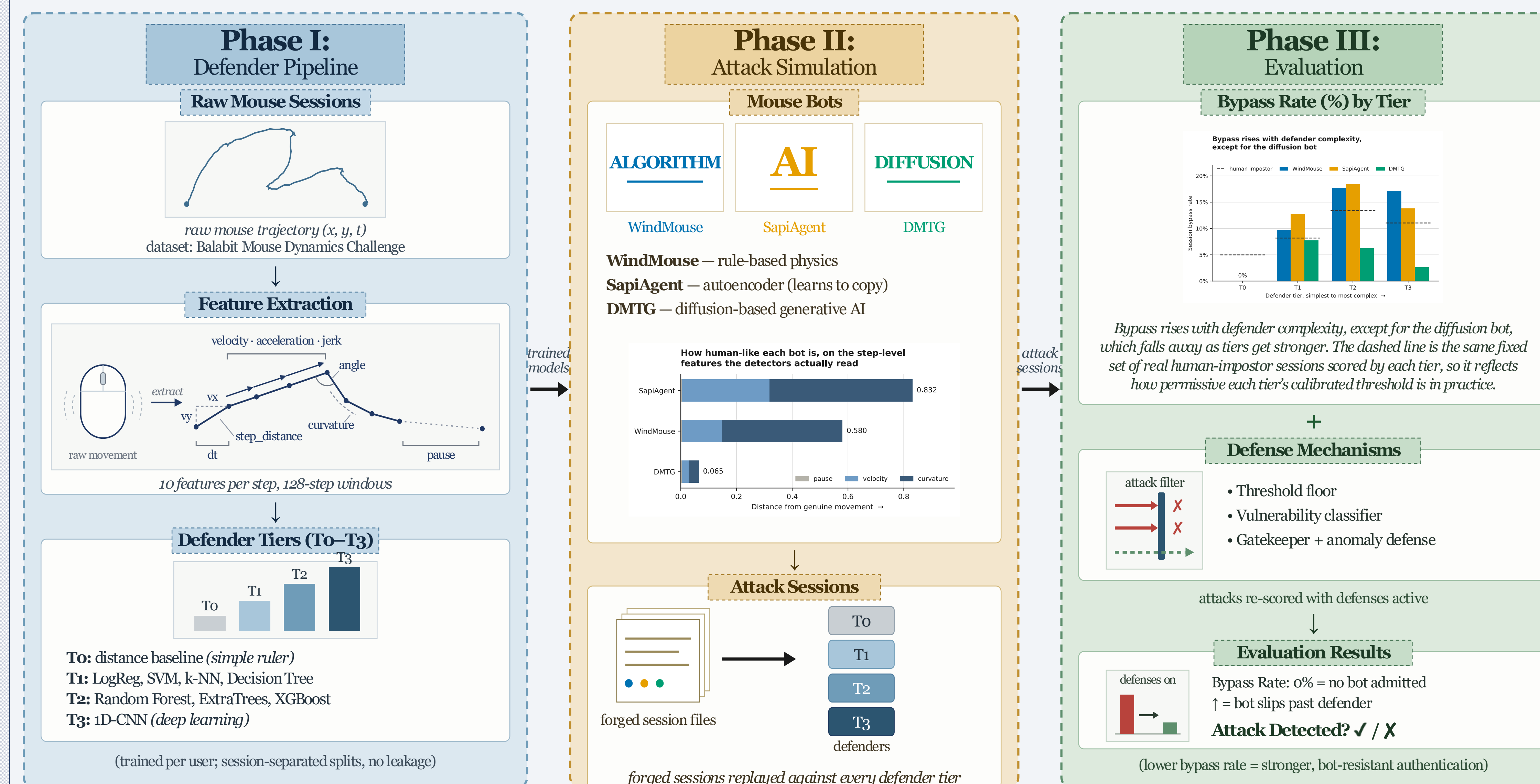


Figure 2: Behavioral authentication pipeline — defender training, synthetic attack simulation, and robustness assessment.

## Results

### Only two of three bots beat a human impostor

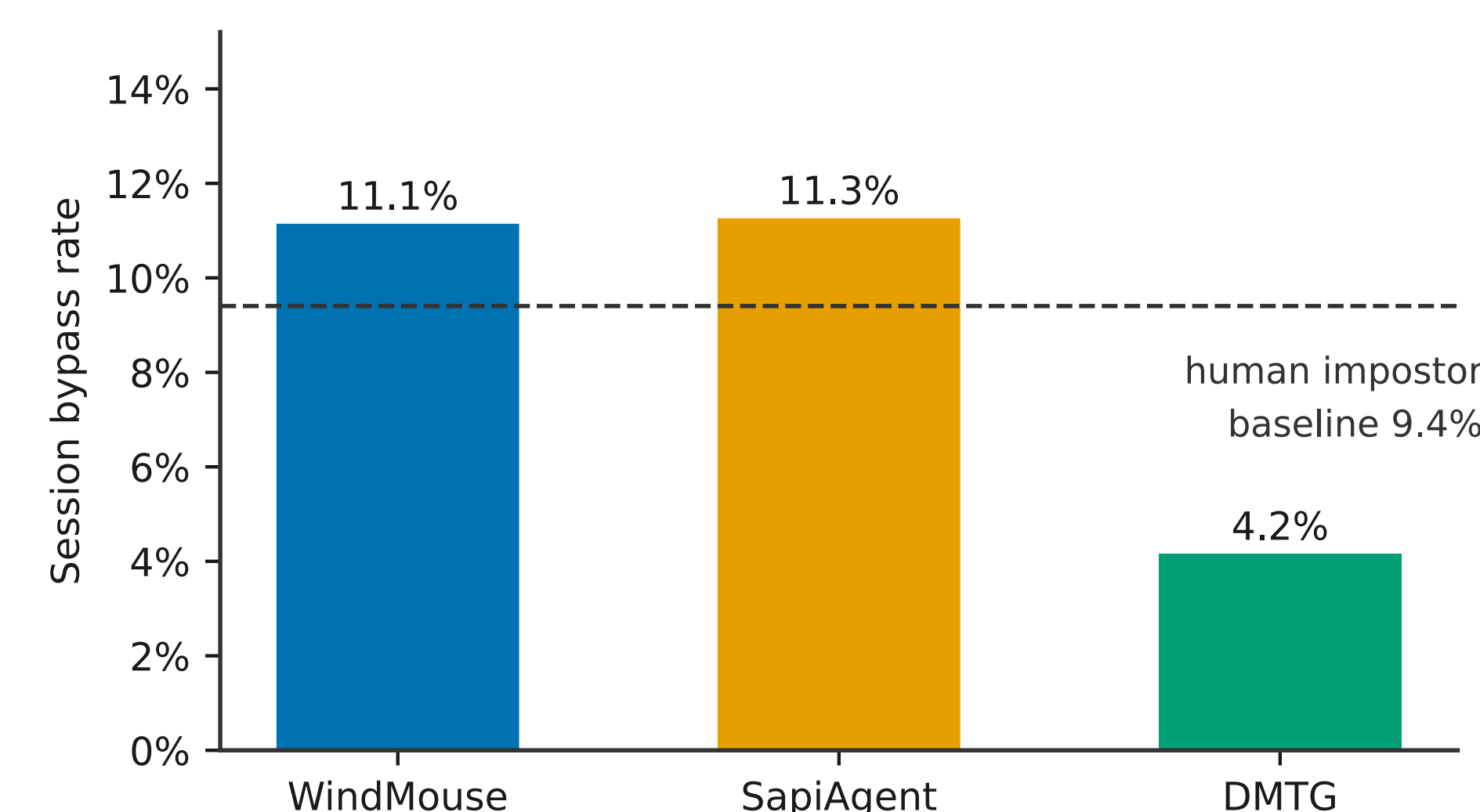


Figure 3. Session bypass rate by attacker, averaged over the four defender tiers, against the real human-impostor baseline of 9.4%. Threshold policy: calibration FAR  $\leq 0.10$ , no threshold floor.

### The same user can be immune to one bot and wide open to another

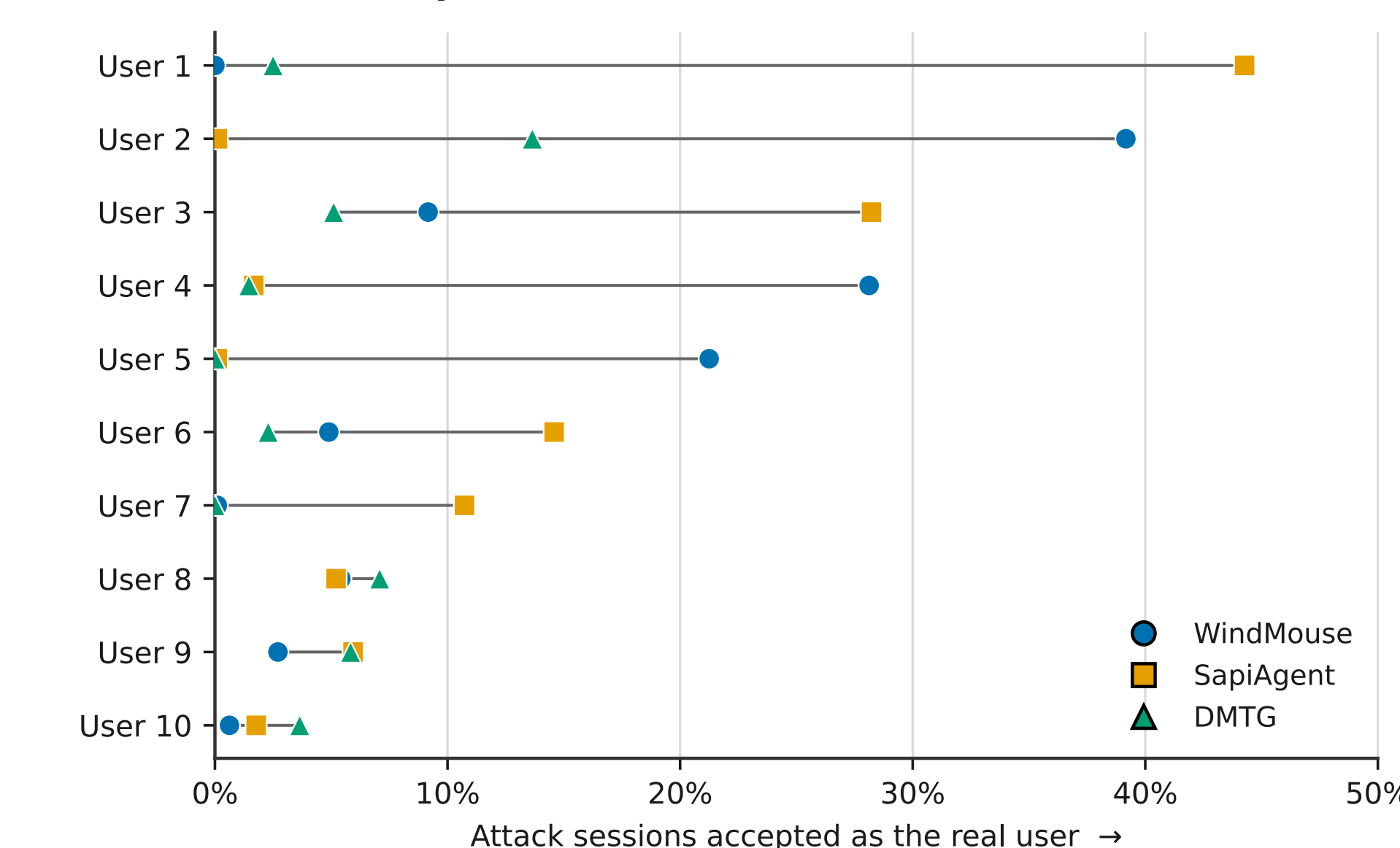


Figure 4. Per-user session bypass by attacker, averaged over defender tiers. Users sorted by worst-case bypass; User 1–10 = Balabit users 21, 15, 29, 23, 9, 7, 35, 20, 16, 12. Vulnerable sets are largely disjoint: an attacker free to choose the best tool per target succeeds 20.3% of the time, nearly double the best single attacker.

### The threshold floor is a targeted fix, not a general improvement

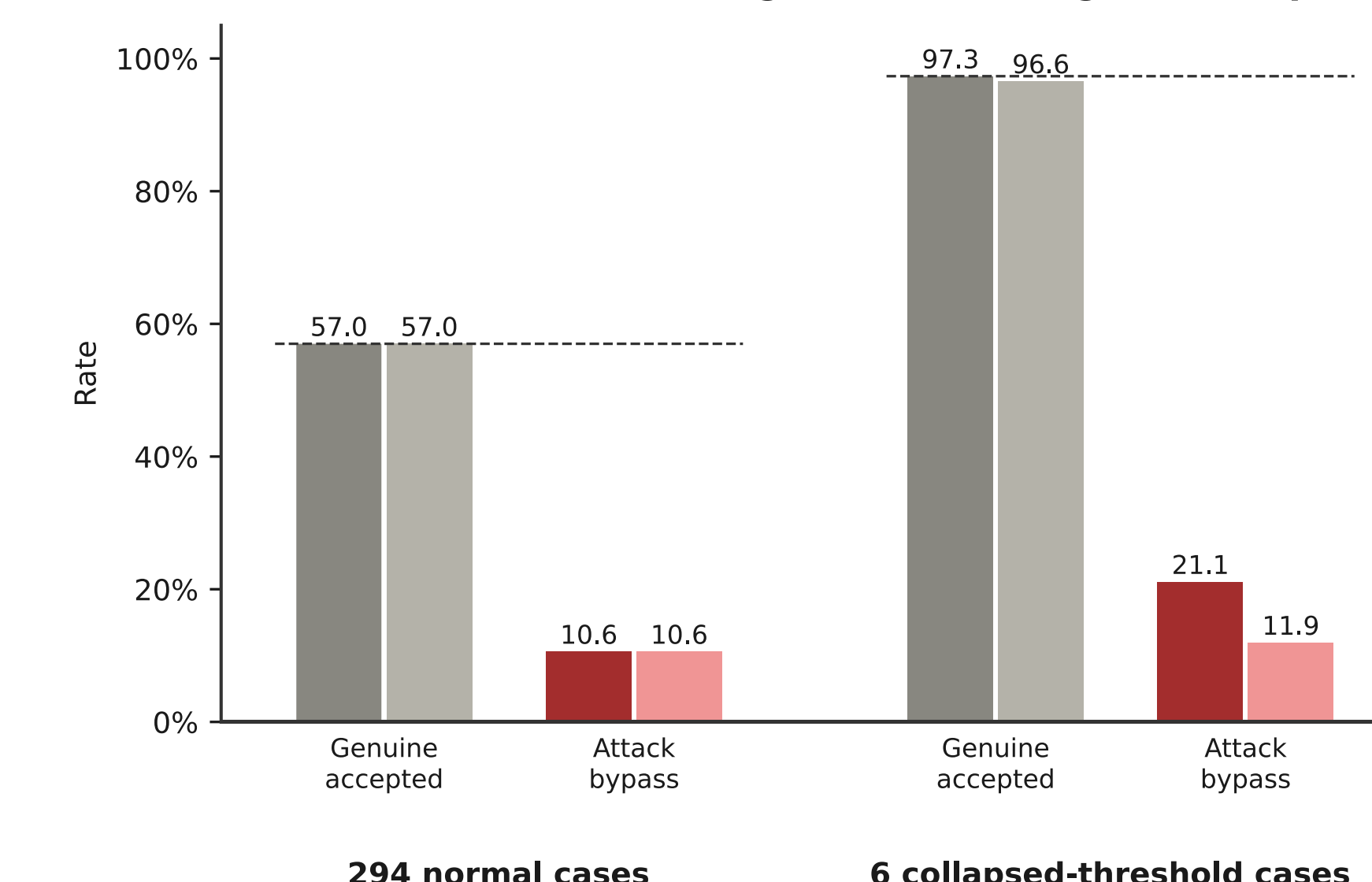


Figure 5. Genuine acceptance and mean attack bypass before and after applying a 0.05 threshold floor, split by whether the calibrated threshold collapsed below 0.05. Case-level means over 300 model-user-seed combinations; 5 of the 6 collapsed cases are T2, one is T3.

### The anomaly layer stops every synthetic session, at a cost to real users

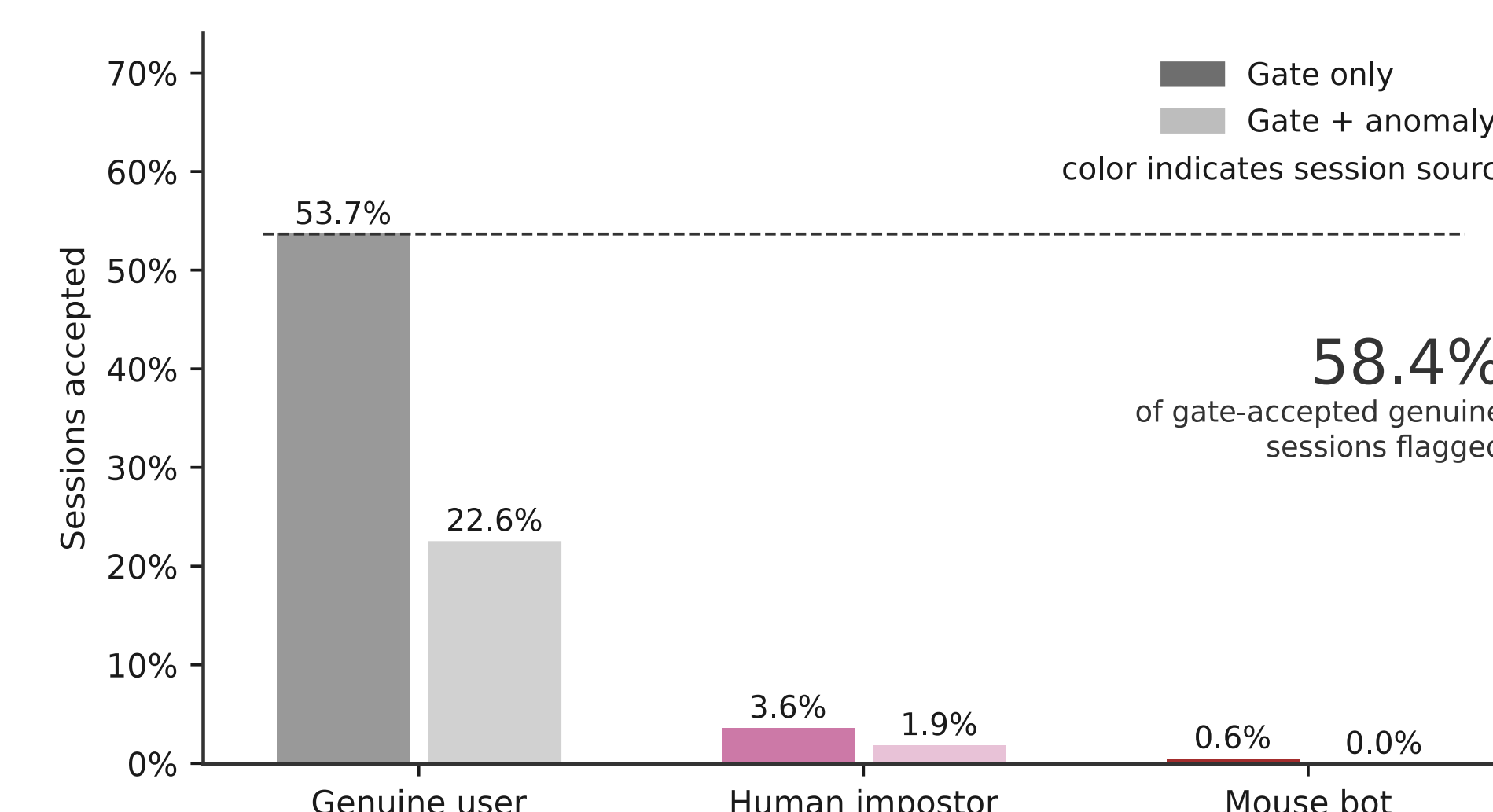


Figure 6. Session acceptance at the gatekeeper (T2 Random Forest) and after the Isolation Forest anomaly layer, by session source. Per-user means.

## Conclusions

- Bots did not outperform a real human impostor.** WindMouse bypassed 11.1% of sessions and SapiAgent 11.3%, barely above the 9.4% real human-impostor baseline. The most sophisticated bot, the diffusion model DMTG, reached only 4.2% — it never cleared the human bar. Sophistication did not translate into threat.
- Vulnerability is user- and bot-specific.** No single bot dominates, and vulnerable sets are largely disjoint: one user blocked WindMouse entirely (0.0%) but fell to SapiAgent 44.3% of the time. An attacker free to choose the best tool per target succeeds 20.3% of the time — nearly double the best single bot.
- The layered defense buys security with usability.** The anomaly layer cut synthetic bypass from 0.56% to 0.00%, but genuine-user acceptance fell from 53.7% to 22.6%, flagging 58.4% of gate-accepted genuine sessions. Every threshold was calibrated for security alone (FAR  $\leq 0.10$ ); calibrating toward usability is untested.
- A human baseline and leakage-free evaluation are what make these numbers interpretable.** Session-separated splits are required for reporting security conclusions, and scoring real human impostors through the same pipeline gives every bypass rate a bar to clear — prior work typically evaluates a single attacker type against a single detector.

## Future Work

- Calibrate toward usability.** Thresholds here targeted security only (FAR  $\leq 0.10$ ); the reverse sweep is untested.
- Build bots that adapt.** All three are target-blind, and the 20.3% best-tool figure assumes a selection step we never implemented.
- Scale beyond ten users** — ten can show *that* vulnerability is user-specific, not *why*.
- Normalize scores across model families** — a raw 0.05 threshold floor is not comparable across SVM, tree, and CNN distributions.
- Extend the diffusion bot with the dropped style-transfer module**, preserving target-blindness, and study entropy and step-count against bypass rate.

## References

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